Rishav Singh

+917016993938 | rsh04613@gmail.com | rishav-singh-0 | blog.rishavs.in

Education

Dharmsinh Desai University **2019 - 2023**
B.Tech. in Electronics Communication Engineering; CGPA - 8.65/10 *Nadiad, Gujarat*

Experience

MediaTek India Technology Pvt. Ltd **June 2025 - Present**
Senior Software Engineer - GPU *Noida, Uttar Pradesh*

- Integrated ARM GPUs across multiple MediaTek Dimensity platforms, covering bring-up, driver adaptation, BSP porting and AOSP framework alignment across multiple Linux Kernel and Android AOSP versions
- Enhance platform stability through security fixes, and coordinated kernel–userspace updates
- Successfully migrated GPU drivers and graphics stacks across multiple OS generations, including the transition from Android 16 to Android 17

Matrix Comsec Pvt. Ltd **Jan 2023 - May 2025**
Embedded Platform Engineer *Vadodara, Gujarat*

- Developed Network Video Recorders and IP Cameras, overseeing BSP creation with Buildroot and crafting kernel drivers for diverse functionalities, collaborating with cross team software architects
- Worked on 104 camera variants using 3 SoCs(SSC339G, SSC335DE, C5-36GW) and 12 NVR variants with 2 SoCs (RK3568, RK3588), ensuring compatibility and reliability
- Solely maintained multiple product lines, improved build times upto 80%, Makefiles, and release processes to enhance efficiency
- Designed drivers for PTZ camera motors, achieving 38x optical zoom with combination of 5 motors, and gyro-based image stabilization
- Resolved critical hardware-software issues, including NVR HDMI signal problems and OEM USB upgrade security

eYantra, IIT Bombay **May 2021 - Jul 2021**
Intern *Mumbai, Maharashtra*

- Optimized auto evaluation tool for working on Linux Kernels
- Created unified platform and connected Django backend with Vue frontend and automated dynamic docker container generation on submission and result evaluation based on each theme requirement

Projects

IoT based Hydroponics(Govt. Granted) | *C++, Python, Flutter, Firebase* **Apr 2022 - Jun 2022**

- A Gujarat government granted(SSIP) project to modify hydroponics based farming system with IoT in order to improve quality of soil less farming
- Connected the hydroponics plant to the cloud server for data storage and control through flutter android app

Holonomic Drive Robot | *ESP32, Atmega2560, ROS* **Jan 2023 - Mar 2023**

- Built a three-wheeled holonomic robot for precision path planning and image-based drawing
- Developed algorithms for motion control and image contour processing, enabling intricate pattern creation

Vitarana Drone - Autonomous Drone Based Delivery System | *Python, ROS, Gazebo, Git* **Oct 2020 - Feb 2021**

- Used Robot Operating System(ROS) for Autonomous Drone based delivery system in a simulated environment
- Learned about team management and task division and distribution for achieving larger goals

Technical Skills

Languages: C, Python, Bash, Go
Technologies: Buildroot, Busybox, Rockchip, Innofusion SoCs
Tools/Frameworks: Linux, Git, GitHub, SVN, Jira, Docker
Fields: BSP Development, Board Bringup, Linux Device Drivers, IPCamera and NVR Development

Achievements

- **Led** the team to pre-finals round in eYantra Robotics Competetion and were in **top 20** among **2603 teams of 572 colleges** all over the globe
- Winner of Code-e-Fest Competition in DDU annual festival (2021)